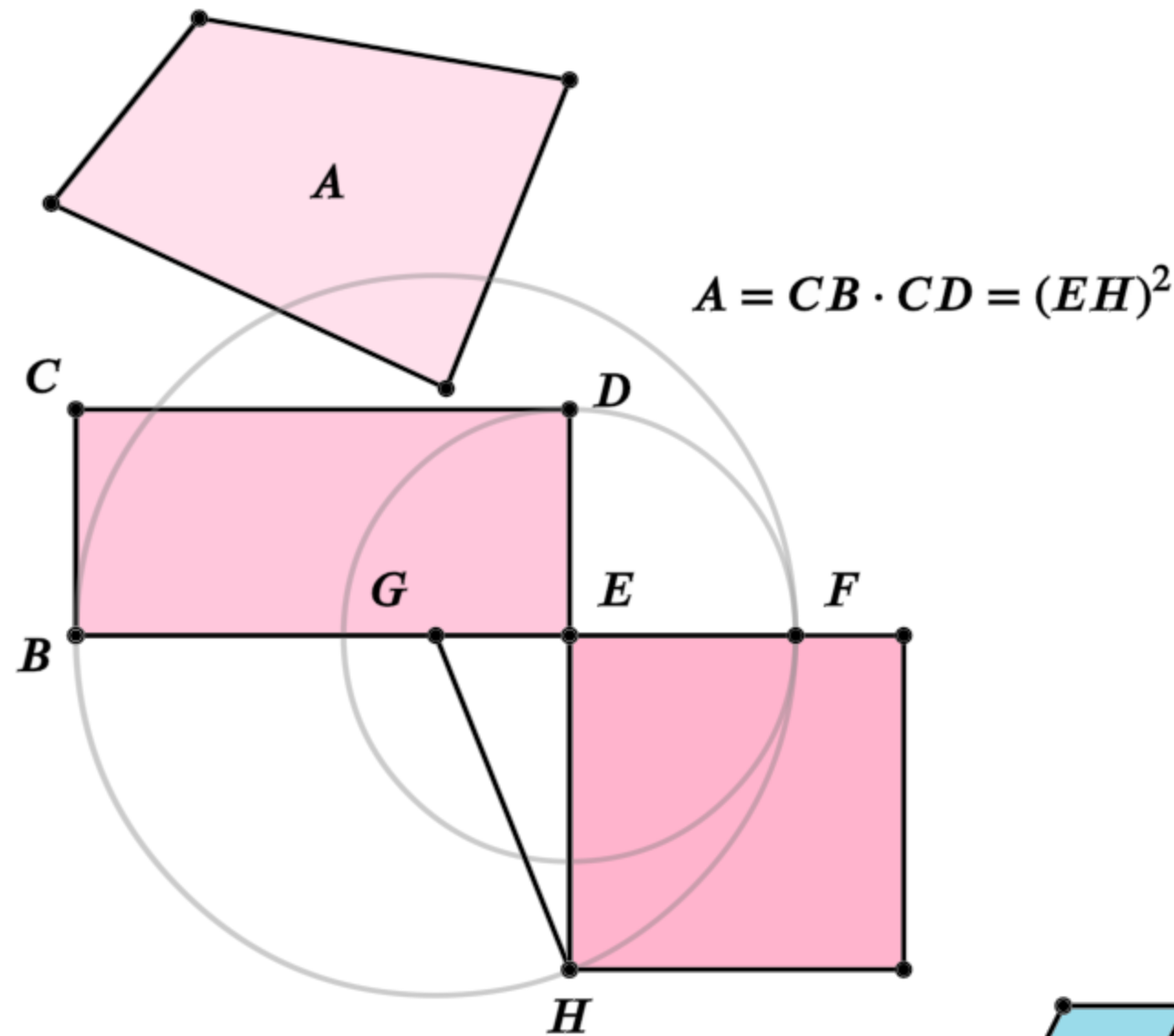


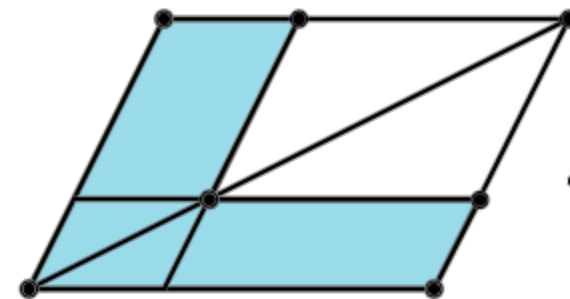
Euclid's Elements

Book II



It is a remarkable fact in the history of geometry, that the Elements of Euclid, written two thousand years ago, are still regarded by many as the best introduction to the mathematical sciences.

- Florian Cajori,
A History of Mathematics (1893)



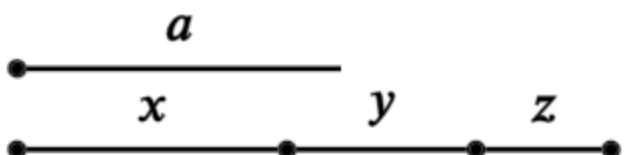
Definitions:

Any rectangular parallelogram is said to be contained by the two straight lines containing the right angle.

And in any parallelogrammic area let any one whatever of the parallelograms about its diameter with the two complements be called a gnomon.

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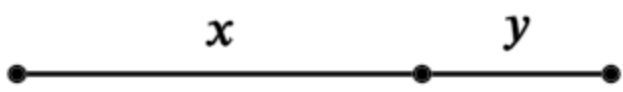
1.


$$a(x + y + z) = ax + ay + az$$

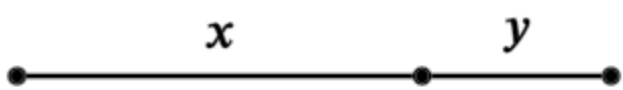
2.


$$(x + y)^2 = x(x + y) + y(x + y)$$

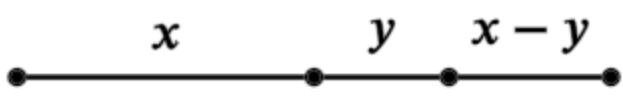
3.


$$y(x + y) = yx + y^2$$

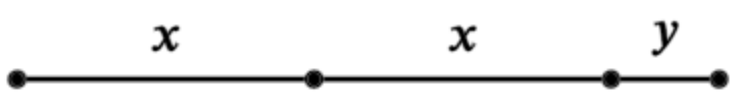
4.


$$(x + y)^2 = x^2 + y^2 + 2xy$$

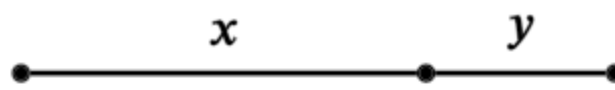
5.


$$(x + y)(x - y) + y^2 = x^2$$

6.


$$y(2x + y) + x^2 = (x + y)^2$$

7.


$$(x + y)^2 + y^2 = x^2 + 2y(x + y)$$

8.


$$(x + 2y)^2 = 4y(x + y) + x^2$$

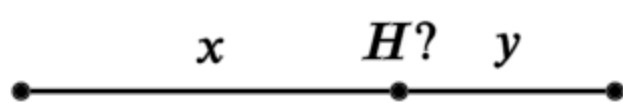
9.


$$(x + y)^2 + (x - y)^2 = 2(x^2 + y^2)$$

10.

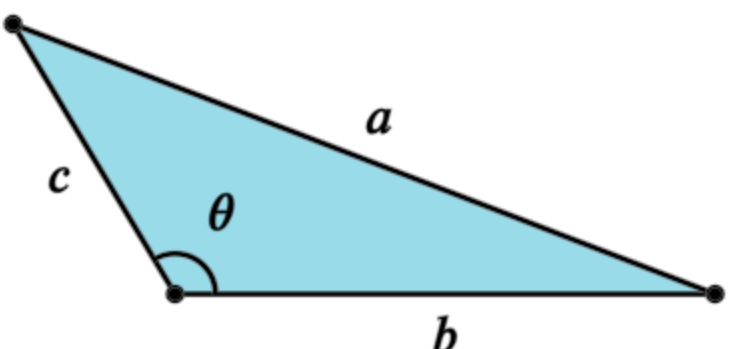

$$(2x + y)^2 + y^2 = 2(x^2 + (x + y)^2)$$

11.



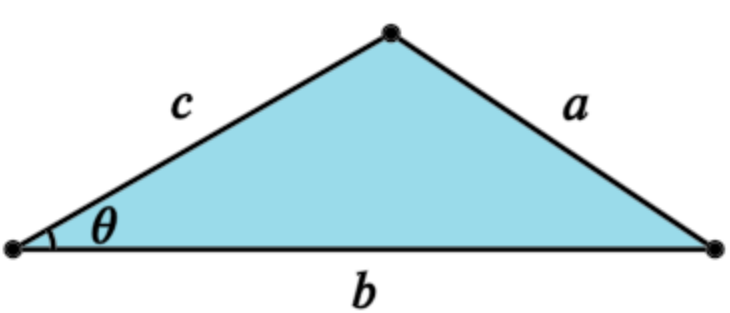
Find H such that: $y(x + y) = x^2$

12.



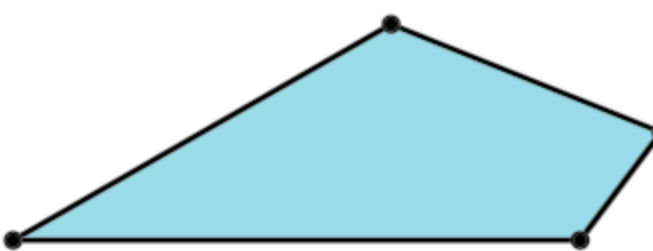
Cosine Law, $a^2 = b^2 + c^2 - 2bc \cos \theta$

13.



Cosine Law. $a^2 = b^2 + c^2 - 2bc \cos \theta$

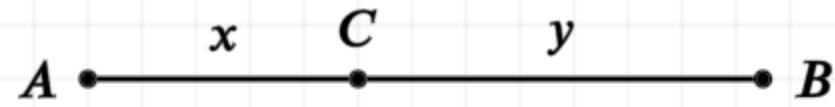
14.



Construct a square equal to the polygon

Proposition 2 of Book 2

If a straight line be cut at random, the rectangles contained by the whole and both of the segments are equal to the square on the whole



In other words

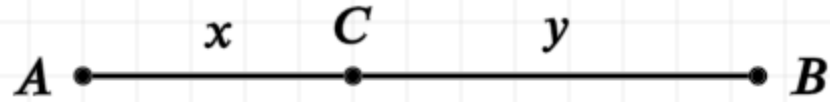
Let AB be a straight line, arbitrarily cut at point C

$$AB = AC + CB$$

$$a = x + y$$

Proposition 2 of Book 2

If a straight line be cut at random, the rectangles contained by the whole and both of the segments are equal to the square on the whole



In other words

Let AB be a straight line, arbitrarily cut at point C

Then the area of the square formed by line AB is equal in area to the sum of the rectangles formed by line AB and AC , and line AB and BC

$$AB = AC + CB$$

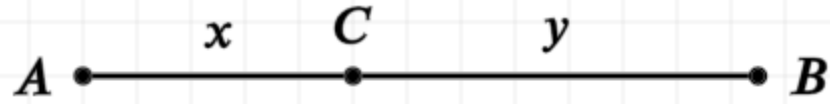
$$AB^2 = AB \cdot AC + AB \cdot BC$$

$$a = x + y$$

$$(x + y)^2 = (x + y)x + (x + y)y$$

Proposition 2 of Book 2

If a straight line be cut at random, the rectangles contained by the whole and both of the segments are equal to the square on the whole



In other words

Let AB be a straight line, arbitrarily cut at point C

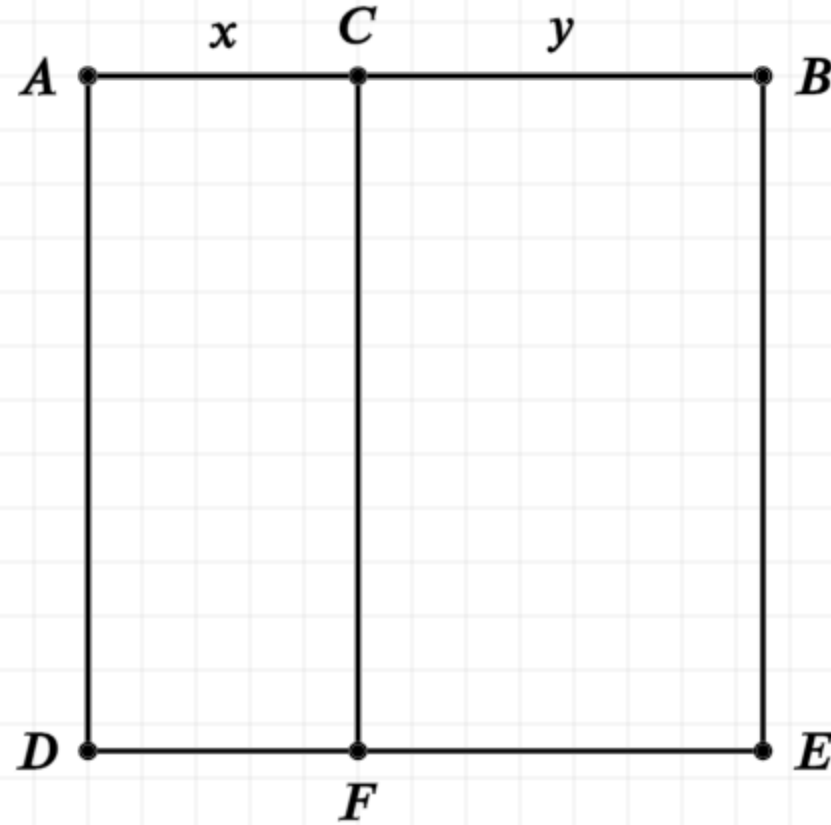
$$AB^2 = AB \cdot AC + AB \cdot BC$$

Proof:

$$AB = AC + CB$$

Proposition 2 of Book 2

If a straight line be cut at random, the rectangles contained by the whole and both of the segments are equal to the square on the whole



$$AB = AC + CB$$

$$AB = AD = CF$$

In other words

Let AB be a straight line, arbitrarily cut at point C

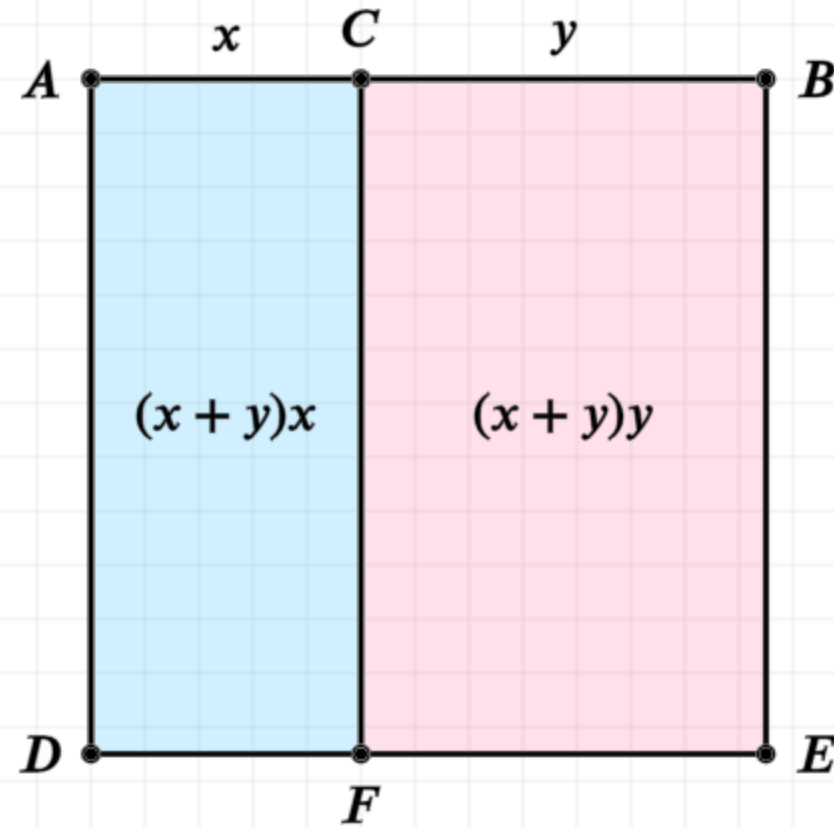
$$AB^2 = AB \cdot AC + AB \cdot BC$$

Proof:

Draw a square ABED on the line AB (I.46) and draw a line CF parallel to either AD or BE (I.31)

Proposition 2 of Book 2

If a straight line be cut at random, the rectangles contained by the whole and both of the segments are equal to the square on the whole



$$AB = AC + CB$$

$$AB = AD = CF$$

$$\text{rectangle } AE = \text{rectangle } AF + \text{rectangle } CE$$

In other words

Let AB be a straight line, arbitrarily cut at point C

$$AB^2 = AB \cdot AC + AB \cdot BC$$

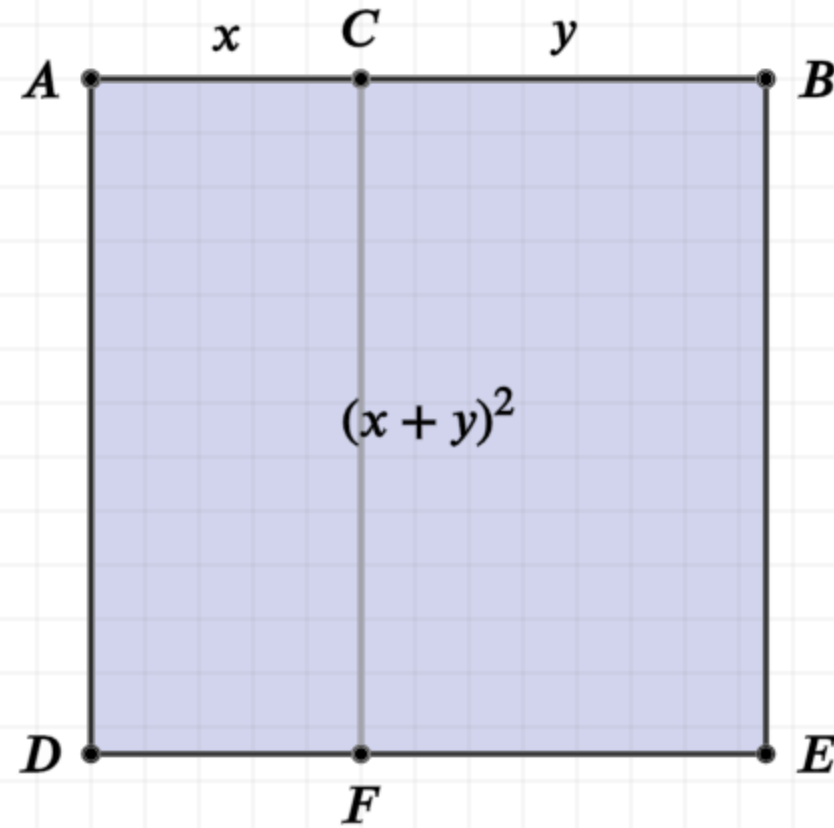
Proof:

Draw a square ABED on the line AB (I.46) and draw a line CF parallel to either AD or BE (I.31)

The rectangle AE is the sum of the rectangles AF and CE

Proposition 2 of Book 2

If a straight line be cut at random, the rectangles contained by the whole and both of the segments are equal to the square on the whole



$$AB = AC + CB$$

$$AB = AD = CF$$

$$AE = AF + CE$$

$$AE = AB \cdot AD, \quad \therefore AE = AB^2$$

In other words

Let AB be a straight line, arbitrarily cut at point C

$$AB^2 = AB \cdot AC + AB \cdot BC$$

Proof:

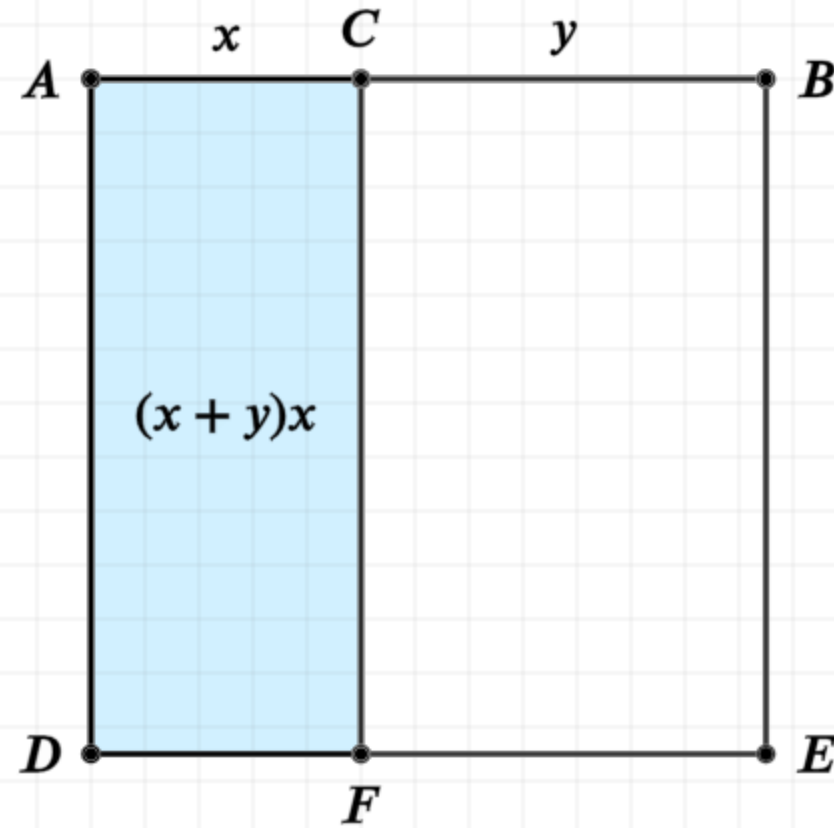
Draw a square ABED on the line AB (I.46) and draw a line CF parallel to either AD or BE (I.31)

The rectangle AE is the sum of the rectangles AF and CE

Since AD is equal in length to AB, the rectangle AE is equal to the square contained by line AB

Proposition 2 of Book 2

If a straight line be cut at random, the rectangles contained by the whole and both of the segments are equal to the square on the whole



$$AB = AC + CB$$

$$AB = AD = CF$$

$$\text{rectangle } AE = \text{rectangle } AF + \text{rectangle } CE$$

$$\text{rectangle } AE = AB \cdot AD, \quad \therefore \text{rectangle } AE = AB^2$$

$$\text{rectangle } AF = AD \cdot AC, \quad \therefore \text{rectangle } AF = AB \cdot AC$$

In other words

Let AB be a straight line, arbitrarily cut at point C

$$AB^2 = AB \cdot AC + AB \cdot BC$$

Proof:

Draw a square ABED on the line AB (I.46) and draw a line CF parallel to either AD or BE (I.31)

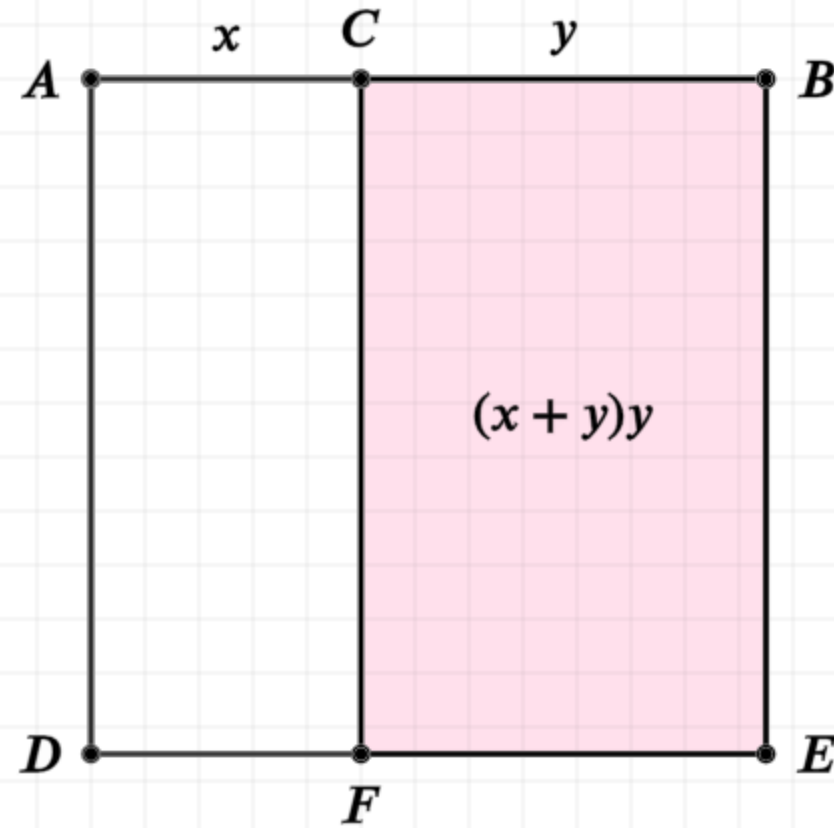
The rectangle AE is the sum of the rectangles AF and CE

Since AD is equal in length to AB, the rectangle AE is equal to the square contained by line AB

Similarly, the rectangle AF is equal to the rectangle contained by lines AB and AC

Proposition 2 of Book 2

If a straight line be cut at random, the rectangles contained by the whole and both of the segments are equal to the square on the whole



$$AB = AC + CB$$

$$AB = AD = CF$$

$$AE = AF + CE$$

$$AE = AB \cdot AD, \quad \therefore AE = AB^2$$

$$AF = AD \cdot AC, \quad \therefore AF = AB \cdot AC$$

$$CE = CF \cdot CB, \quad \therefore CE = AB \cdot CB$$

In other words

Let AB be a straight line, arbitrarily cut at point C

$$AB^2 = AB \cdot AC + AB \cdot BC$$

Proof:

Draw a square ABED on the line AB (I.46) and draw a line CF parallel to either AD or BE (I.31)

The rectangle AE is the sum of the rectangles AF and CE

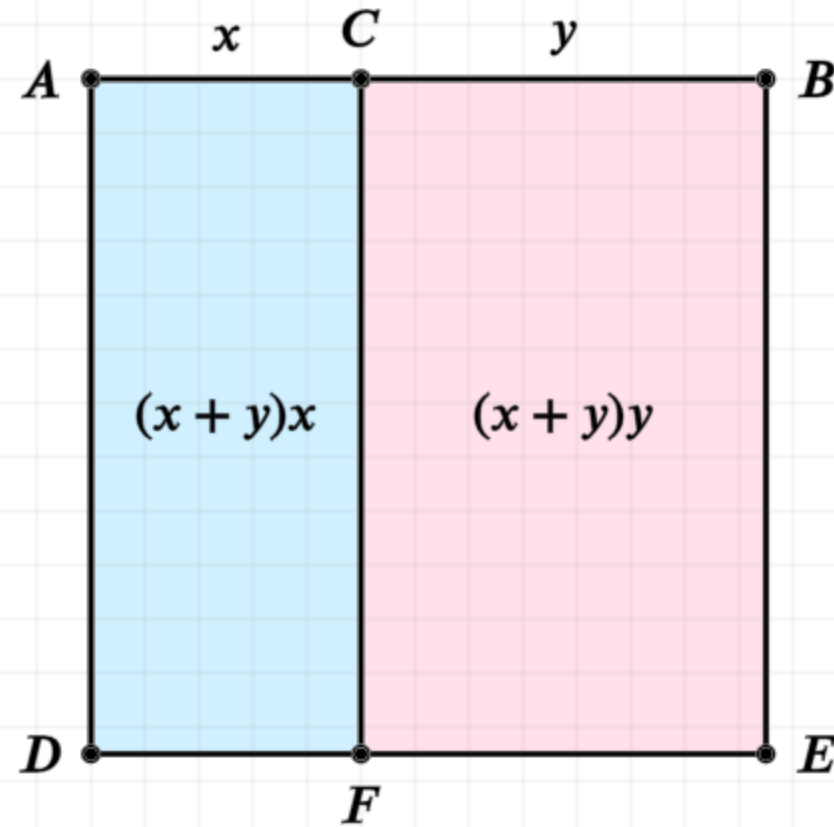
Since AD is equal in length to AB, the rectangle AE is equal to the square contained by line AB

Similarly, the rectangle AF is equal to the rectangle contained by lines AB and AC

Since AB equals CF (I.34), CE is equal to the rectangle contained by lines AB and CB

Proposition 2 of Book 2

If a straight line be cut at random, the rectangles contained by the whole and both of the segments are equal to the square on the whole



$$AB = AC + CB$$

$$AB = AD = CF$$

$$\text{Area } AE = \text{Area } AF + \text{Area } CE$$

$$\text{Area } AE = AB \cdot AD, \quad \therefore \text{Area } AE = AB^2$$

$$\text{Area } AF = AD \cdot AC, \quad \therefore \text{Area } AF = AB \cdot AC$$

$$\text{Area } CE = CF \cdot CB, \quad \therefore \text{Area } CE = AB \cdot CB$$

$$AB^2 = AB \cdot AC + AB \cdot CB$$

In other words

Let AB be a straight line, arbitrarily cut at point C

$$AB^2 = AB \cdot AC + AB \cdot BC$$

Proof:

Draw a square ABED on the line AB (I.46) and draw a line CF parallel to either AD or BE (I.31)

The rectangle AE is the sum of the rectangles AF and CE

Since AD is equal in length to AB, the rectangle AE is equal to the square contained by line AB

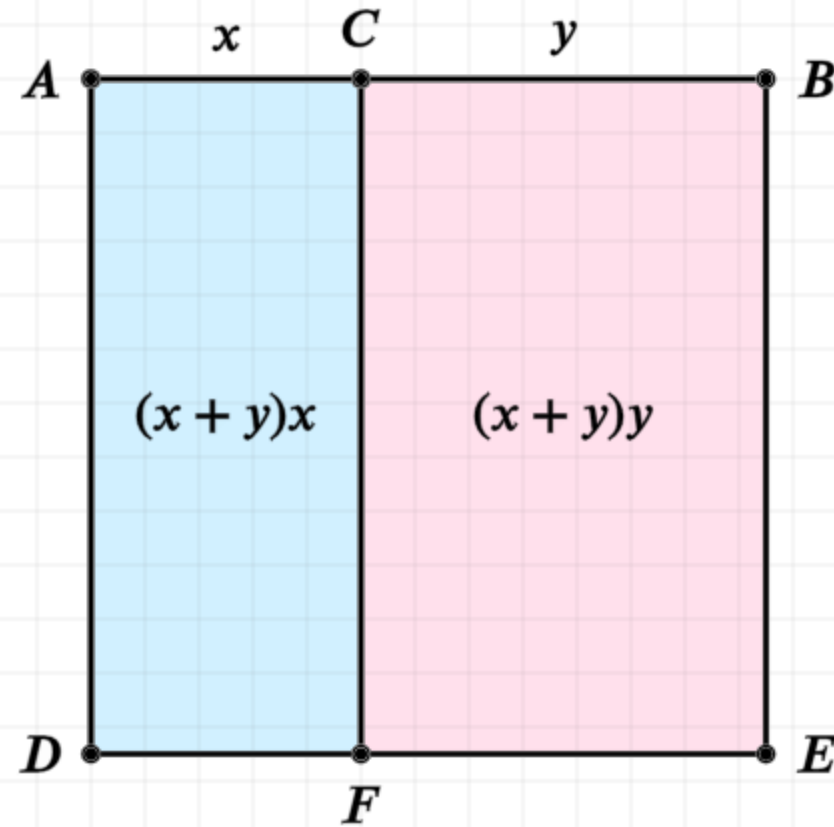
Similarly, the rectangle AF is equal to the rectangle contained by lines AB and AC

Since AB equals CF (I.34), CE is equal to the rectangle contained by lines AB and CB

Thus the square of AB is equal to the rectangle formed by AB, AC and the rectangle formed by AB, CB

Proposition 2 of Book 2

If a straight line be cut at random, the rectangles contained by the whole and both of the segments are equal to the square on the whole



$$AB = AC + CB$$

$$AB = AD = CF$$

$$AE = AF + CE$$

$$AE = AB \cdot AD, \quad \therefore AE = AB^2$$

$$AF = AD \cdot AC, \quad \therefore AF = AB \cdot AC$$

$$CE = CF \cdot CB, \quad \therefore CE = AB \cdot CB$$

$$AB^2 = AB \cdot AC + AB \cdot CB, \quad (x + y)^2 = (x + y)x + (x + y)y$$

In other words

Let AB be a straight line, arbitrarily cut at point C

$$AB^2 = AB \cdot AC + AB \cdot BC$$

Proof:

Draw a square ABED on the line AB (I.46) and draw a line CF parallel to either AD or BE (I.31)

The rectangle AE is the sum of the rectangles AF and CE

Since AD is equal in length to AB, the rectangle AE is equal to the square contained by line AB

Similarly, the rectangle AF is equal to the rectangle contained by lines AB and AC

Since AB equals CF (I.34), CE is equal to the rectangle contained by lines AB and CB

Thus the square of AB is equal to the rectangle formed by AB, AC and the rectangle formed by AB, CB

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